

```
Download:
wget http://downloads.asterisk.org/pub/telephony/libpri/
libpri-1.4.15.tar.gz
Unzip:
tar xzvf libpri-1.4.15.tar.gz
Install:
cd /usr/src/libpri-1.4.15.tar.gz
make
make install
```

Installing Asterisk

Now, we are ready to install a basic version of Asterisk. Let's download version 11, which is a LTS (Long Term Support) release.

```
Download:
wget http://downloads.asterisk.org/pub/telephony/asterisk/
releases/asterisk-11.11.0.tar.gz
Unzip:
tar xzvf asterisk-11.11.0.tar.gz
Install:
cd /usr/src/asterisk-11.11.0
./configure
make menuselect
```

Now you get a GUI, a *Menuselect* menu, listing categories such as applications, channel drivers, and PBX modules. On the right-hand side, you'll see a list of modules that correspond with the select category. At the bottom of the screen, you'll see two buttons. You can use the *Tab* key to switch between the various sections, and press the *Enter* key to select or unselect a particular module. If you see *[*]* next to a module name, it signifies that the module has been selected. If you see **XXX* next to a module name, it signifies that the select module cannot be built, as one of its dependencies is missing. In that case, you can look at the bottom of the screen for the line labelled 'Depends upon:' for a description of the missing dependency. Once *Menuselect* is executed, and the required applications, channel drivers and modules selected, exit the GUI.

```
make
make install
make config
make samples
```

make samples installs the configuration and contains more than just an example configuration. The sample configuration files historically were used predominantly for documentation of available options. As such, they contain many examples for configuring Asterisk that may not be ideal for standard deployments. Sample installation is helpful as it creates sample configurations of all configuration files. We can select specific ones like *sip.conf* and *extensions.conf* and overwrite them. Else, we might need to create a series of configuration files from scratch.

Start services

```
service dahdi start
service asterisk start
```

For Debian or Ubuntu users, the installation of basic libraries and start services is slightly different, as shown below:

```
apt-get install build-essential wget libssl-dev libncurses5-
dev libnewt-dev libxml2-dev linux-headers-$(uname -r)
libsqlite3-dev uuid-dev
Start services:
/etc/init.d/dahdi start
/etc/init.d/asterisk start
```

All the other steps remain the same.

Installing SIP phones

Basic Asterisk is an IP PBX, which can interconnect IP phones with the SIP (Session Initiation Protocol) or IAX (Inter Asterisk eXchange) protocol. SIP is quite widespread and best suited for communication with other systems. IAX is a very compact protocol with less bandwidth requirements, which is proprietary to Asterisk, and can be used where limited bandwidth is available.

There are three types of endpoints you would typically provide your users with, which could serve as a telephone set. They are popularly referred to as hardphones — physical devices like normal telephones with a handset, buttons, etc; softphones — software applications that run on top of laptops or desktops; and analogue terminal adaptors (ATAs) — connectors for traditional analogue devices like analogue phones, faxes, etc. All of them are configured as SIP extensions in Asterisk. To configure the SIP extensions, we need to edit */etc/asterisk/sip.conf*.

```
[general]
context=unauthenticated
allowguest=no
alwaysauthreject=yes
```

```
[test-phone]()
type=friend
host=dynamic
context=office-device
disallow=all
allow=ulaw
allow=alaw
```

```
[ramesh](test-phone)
secret=mySUP3r@ss1
```

```
[suresh](test-phone)
secret=mySUP3r@ssII
```